THE UNITED REPUBLIC OF TANZANIA

DODOMA REGION

FORM FOUR MOCK- EXAMINATION

032/1 **CHEMISTRY 1**

Time: 3 Hours **August, 2023**

Instructions

1. This paper consists of section A, B and C with total of eleven (11) questions
2. Answer all questions in section A and B and two (2) question from section C
3. Cellular phones and any unauthorised material are not allowed in the examination room
4. Write your Examination Number on every page of your answer booklet(s)
5. Non programmable calculator may be used
6. The following constant may be used
   Atomic masse:
   H=1, C=12, N=14, O=16, Na=23, S= 32, Ca=40, Cl= 35.5, Cu= 64 and Zn=65
   Avogadro's number = 6.02×10^{23}
   GMV at s.t.p = 22. 4 dm^3
   1 faraday = 96,500 coulombs
   Standard pressure = 760 mm Hg
   Starndard tempreture = 273K
   1 litre = 1 dm^3= 1000cm^3

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SECTION A: (16 marks)

Answer all questions in this section

1. For each of the items (i)-(x) choose the correct answer from given alternatives and write its letter beside item number in the answer booklet provided

i. Which of the following is not a use of a solvent?
   A. Bleaching agent
   B. Greasing.
   C. Stain removal.
   D. Cleaning
   E. Universal solvent

ii. In which step of scientific procedures does the hypothesis is either proved or disproved?
   A. Data collection and analysis.
   B. Experimentation.
   C. Data interpretation
   D. Formulation of hypothesis.
   E. Conclusion

iii. iii) Which of the following groups consist of home Care products?
   A. Yeast, plastic and disinfectant.
   B. Clothes, soap and stone.
   C. Air freshener, detergent and antiseptic.
   D. Petrol, air freshener and paints.
   E. Air freshener, detergent and disinfectant

iv. An electric current of 0.2A was passed through an electrolyte for 16.67minutes. The quantity of electricity passed is;

   A. 200.04cuolombs
   B. 2000.004cuolombs
   C. 1000cuolombs
   D. 0.254cuolombs
   E. 0.00789culombs

v. What are the main components of crude oil?
   A. Gases, liquids and solids.
   B. Gases, water and diesel.
   C. Mineral, salts and oils
   D. Gasoline, hydrogen and milk.
   E. Methane only

vi. What will happen when sulphur dioxide is mixed with potassium dichromate (iv) solution in Presence of sulphuric acid?
   A. Sulphur will be formed.

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B. Green chromate (iii) sulphate will be formed
C. Blue water will be formed.
D. Sulphur dioxide gas will be formed
E. Orange potassium sulphate will be formed

vii. Which of the following reaction is a most convenient method of producing sulphur dioxide during contact process
A. Iron (ii) sulphide react with oxygen to form sulphur dioxide and iron(iii) oxide
B. Zinc sulphide reacts with oxygen to form sulphur dioxide and zinc oxide
C. Sulphur reacts with oxygen to form sulphur dioxide
D. Oxygen reacts with hydrogen to form water
E. Sulphur dioxide reacts with oxygen to form sulphur trioxide

viii. The copper (ii) oxide reacts with hydrogen gas to form copper metal and water. What will be the mass of reduced element?
A. 4g.
B. 64g.
C. 18g.
D. 80g.
E. 40g

ix. During the steam reforming method in industrial preparation of hydrogen, the steam reacts with what compound to produce hydrogen gas?
A. Water.
B. Carbon monoxide.
C. Methane.
D. Sulphur dioxide.
E. Oxide

x. What will be the simplest test for the hardness of water?
A. Shaking water with chalk.
B. Mixing water with soap less detergent.
C. Formation of scum.
D. Shaking water with soap solution.
E. Formation of dolomite

2. Match the colour formed in LIST A with the corresponding metal burning in air.

LIST A	LIST B
i) Burns with a golden yellow flame	A. Lead
ii) Bluish green flame	B. Calcium
iii) Melts easily but show no definite colour flame	C. Zinc
	D. Iron

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iv) Burns with yellow sparks flame v) Burns with a brick red flame V. Pale blue flame	E. Potassium F. Magnesium G. Copper H. Sodium

SECTION B (54marks)

Answer all questions in this section

3. (a) Students are advised to use a non- luminous flame for heating in the laboratory
   (i) Explain how a Bunsen burner produces a non- luminous flame
   (ii) Give a reason as to why advice above given to students.
   (iii) What are the functions of the air holes and barrel in the Bunsen burner?

   (b)Why hydrogen peroxide preferred to potassium chlorate during preparation of oxygen gas?

   (c) Why iron is not usually recommended in construction of steam pipes and boilers?

   (d) What would happen to a well stoppered bottle full of water left in deep freezer over night? Why does this happen.

4. People of Uweto region were complaining about the water they are using in everyday activities that contain different contaminants and pollutants like bad smell, coloured matter, floating materials and other different solid substances. The Uweto people wish their water to be treated and purified but they do not know what to do in treating and purifying water. As a Chemist help Uweto people on how to treat and purify their urban water

5. (a) Explain why carboxylic acid has high melting point and boiling point compared to their corresponding alcohol?

   (b) With help of chemical equation, state two chemical properties of carboxylic acid.

   (c) Explain why does fractional distillation of crude oil gives different products

   (d) (i) Define the term hydrocarbons.
   (ii) Write down the molecular structure and I.U.P.A.C names of the isomers whose molecular formula is C_4H_{10}.

6. (a) In electrolytic production of hydrogen gas, dilute mineral acid is used. Which Method is used in its collection? Give a reason.

   (b) Explain the chemical preference of decorating a copper necklace with silver metal by using electrolysis method

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(c) During electrolysis of brine, sodium was deposited at cathode and chlorine gas released at anode. If 2.0g of sodium were collected at cathode; find the volume of chlorine gas at s.t.p.

7. (a) Neutralization is applied in various useful situations with the aid of balanced chemical equation where necessary; describe any four usefulness of neutralization.

(b) 2.91 g of a monobasic acid, HX, were dissolved in water and made up to 250 cm^3 with water. This solution was titrated with 0.108 M sodium hydroxide solution. 25 cm^3 of the sodium hydroxide solution required 22.5 cm^3 of the HX solution for complete neutralisation. The equation for the reaction is.

$HX_{(aq)} + NaOH_{(aq)} \rightarrow NaX_{(aq)} + H_2O_{(l)}$

(a) Calculate the concentration in (i) g l^{-1} (ii) mol l^{-1} of the acid.

(b) Calculate the molar mass of HX.

8. (a) Gas A was prepared in the laboratory by isolating it from atmospheric air. During it's preparation air was allowed to pass through sodium hydroxide then over heated copper metal.

(i) Identify gas A.

(ii) By using equation, explain what happened when gas A passed through sodium hydroxide and in heated copper metal.

(iii) Write two uses of gas A

(b) (i) What will happen when copper is strongly heated in air at higher temperature and lower temperature respectively.

(ii) Explain how the pure copper is obtained from its ore?

SECTION C (30MARKS)

Answer only two questions

9. (a) Explain the effects of the following to the environment:

(i) Acid rain

(ii) Artificial fertilizers

(iii) Plastic containers

(b) (i) What is the importance of ozone layer in the earth's atmosphere?

(ii) Explain the effect of destroying the ozone layer.

(iii) Why does the increase of the concentration of carbon dioxide gas in the atmosphere results into increase of the earth's surface temperature?

(iv) What control measures should be taken to prevent the destruction of ozone layer? Give three (3) points

(c) (i) Define: " green house effect"

(ii) Give at least two (2) examples of gases which cause green house effect.

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10. There are various environmental plans to be used in controlling the liquid and solid wastes in municipalities. Advise accordingly the Effective plans to be taken in order to achieve such activities. Give six points.
11. (a) Addition of inorganic fertilizers in the farm is not as important as addition of organic manure. Discuss the correctness of this statement in four (4) points

    (b) Soil fertility and soil productivity are mistakenly used to mean the same concept. How do they differ from each other? Give five points

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